Lab 05: Web Architecture and Protocols

INFO 4617/5617 · Week 5 · Chapter 5, Web Architecture and Protocols

Today we code together. Each step below says what we're building and why, and its code cell holds only the comments that frame it: we write the code live, in class. Nothing here is turned in.

We follow one page, the chapter's University of Colorado Boulder article, down through the stack: what your browser's developer tools show, the addresses and DNS lookup under the request, the status codes and headers HTTP carries, and then a time series of the article's readers, from a URL we build ourselves. After each step, a 💬 **Discuss** box holds questions to talk over before the next one; click it to open it.

Before class: open this notebook in the `webdata` environment (chapter 1; it includes `dnspython`), have your email address ready for the User-Agent, and have Chrome, Firefox, or Safari open: Steps 1 and 2 happen in its developer tools. In Safari, first turn on **Settings → Advanced → Show features for web developers**.

On Friday, run every cell above where we stopped (**Run → Run All Above Selected Cell**).

> **Instructor key.** Every cell is filled in and was run against the live sites on 2026-10-21, from a cloud machine that reaches the web through a proxy. The pageviews come from January 2026, the chapter's window, so they will match a rerun; headers, DNS answers, request counts, and timings change from run to run. Three of Step 3's outputs show that machine, not a laptop: its local address is 127.0.0.1, its public address is the proxy's, and curl's narration starts with a `CONNECT` to the proxy (and its `200 Connection Established`) before the TLS handshake. Step 3's optional `ping` cell was not run: the machine has no `ping`, and its proxy carries only HTTP. The counts in Step 2's microsprint come from one page load of each page in headless Chromium, not from a person's browser. The lab follows the textbook at commit d9306b7.

Step 0 — Set up (3 minutes)

Import `requests`, `pandas`, and `matplotlib`, and define the User-Agent every request today will send, in the book's form, `WebDataScience/1.0 (your-email@colorado.edu)`, with your own address. Keep the chapter's article in `ARTICLE`, written as Wikipedia writes titles in URLs, with underscores, and its address in `PAGE`.

```
# Import requests, pandas (as pd), and matplotlib.pyplot (as plt)
# Define HEADERS: a dict whose "User-Agent" is "WebDataScience/1.0 (your-email@colorado.edu)",
 ↪  with your email
# Define ARTICLE, "University_of_Colorado_Boulder", and PAGE, its address on en.wikipedia.org;
 ↪  print HEADERS
import requests
import pandas as pd
import matplotlib.pyplot as plt

HEADERS = {"User-Agent": "WebDataScience/1.0 (your-email@colorado.edu)"}
ARTICLE = "University_of_Colorado_Boulder"
PAGE = f"https://en.wikipedia.org/wiki/{ARTICLE}"
print(HEADERS)
```

```
{'User-Agent': 'WebDataScience/1.0 (your-email@colorado.edu)'}
```

💬 Discuss

- In week 2 we wrote this same User-Agent. Who reads it, and what can they do with your email address?
- Everyone in this room will send the same requests at about the same time. Which of week 2's five questions (legality, consent, proportionality, privacy, server impact) does that make

pressing?

Step 1 — The Inspector, then the same element in Python (6 minutes)

Open the article in your browser, right-click its title, and choose **Inspect** (**Inspect Element** in Safari). The Inspector (Chrome and Safari call it **Elements**) jumps to the title's node in the **DOM**, the browser's live tree of the page. Right-click that node and choose **Copy → Copy selector**. Then check in Python that the selector finds the same element in the HTML `requests` gets, with BeautifulSoup's `select_one()`. **Copy XPath** describes the same element in another language, which `select()` doesn't read: try it, and read the error. For more, see the chapter's Browser Developer Tools, "The Inspector Tab".

```
# Paste the selector you copied into SELECTOR. Request PAGE with HEADERS (timeout=30) and print
 ↪  the status code
# Parse response.text with BeautifulSoup ("lxml"), and print the text of
 ↪  soup.select_one(SELECTOR)
# Then pass the XPath you copied, '//*[@id="firstHeading"]', to soup.select(): catch the error
 ↪  and print its first line
from bs4 import BeautifulSoup

SELECTOR = "#firstHeading"
response = requests.get(PAGE, headers=HEADERS, timeout=30)
print(response.status_code)
soup = BeautifulSoup(response.text, "lxml")
print(soup.select_one(SELECTOR).text)

try:
    soup.select('//*[@id="firstHeading"]')
except Exception as error:
    print(type(error).__name__, str(error).splitlines()[0])
```

```
200
University of Colorado Boulder
SelectorSyntaxError Invalid character '/' position 0
```

💬 Discuss

- The Inspector shows the DOM; `requests` gets what View Source shows, as week 1's side-by-side figure did. If the title had been missing from `response.text`, what would that tell you?
- Why is an `id` like `firstHeading` a safer target for a scraper than a class name chosen for styling?
- In week 4 we found tags with `find()` and `find_all()`. When is a copied selector quicker, and when is it more brittle?

Step 2 — The Network tab, and what one request carries (10 minutes)

Open the Network tab and reload the article: it records only while it's open. The status bar counts the requests; the first row is the HTML document, and the rest are what it asked for. Click that first row and read its **Headers**: the request headers your browser sent, and the response headers it got back. `requests` sends only the one request for the document, so compare it with that row. The **Size** column counts bytes on the wire, which are compressed: the `Content-Encoding` response header says how, and `Content-Length` says how many. `len(response.content)` counts them after `requests` has decompressed them. For more, see the chapter's "The Network Tab".

```
# Print the response's Content-Encoding and Content-Length headers, and len(response.content):
 ↪  which is closer to the Size column?
# Print two headers from Wikimedia's cache, x-cache and age (seconds the cached copy has been
 ↪  kept)
# Print response.request.headers, what requests sent. Compare it with your browser's request
 ↪  headers
```

```
for name in ["Content-Encoding", "Content-Length", "x-cache", "age"]:
    print(f"{name}: {response.headers.get(name)}")
print(f"len(response.content): {len(response.content):,} bytes")
print(response.request.headers)
```

```
Content-Encoding: gzip
Content-Length: 121506
x-cache: cp1101 miss, cp1101 hit/2
age: 966
len(response.content): 768,062 bytes
{'User-Agent': 'WebDataScience/1.0 (your-email@colorado.edu)', 'Accept-Encoding': 'gzip,
 ↪  deflate', 'Accept': '*/*', 'Connection': 'keep-alive'}
```

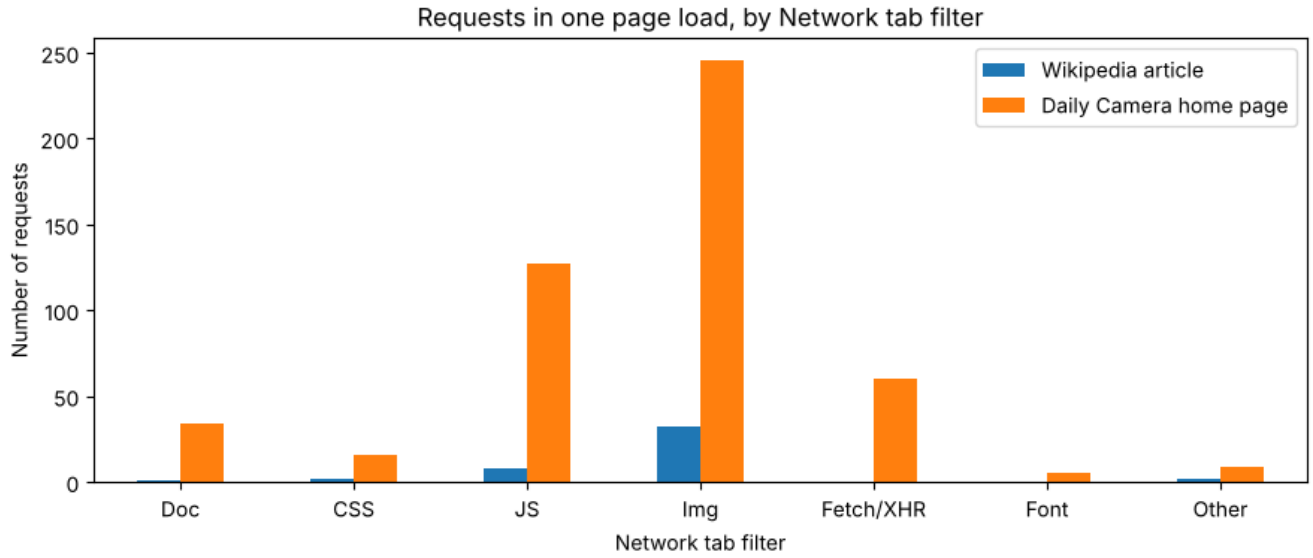
⏱️ Microsprint (5 minutes, alone or in pairs)

Do: a short version of Additional Exercise 1. Reload a news site you read with the Network tab open, and note the total number of requests and how many the **Doc**, **CSS**, **JS**, **Img**, **Fetch/XHR**, and **Font** filters each show (the rest are Other). Do the same for the article. Type the counts into the cell, and plot them side by side. **Share:** the site, its total number of requests, and the share that are scripts (JS).

```
# Microsprint: put your counts in a dict of dicts, one per page: {"Wikipedia article": {"Doc": 1,
 ↪  "JS": 8, ...}, "<your site>": {...}}
# Make a DataFrame from it, and draw a bar chart (one group per filter) with a title and labeled
 ↪  axes
# Print each page's total number of requests, and the share of them that are JS
# The key's counts: one load of each page in headless Chromium on 2026-10-21, sorted into the
 ↪  Network tab's filters
counts = {
    "Wikipedia article": {"Doc": 1, "CSS": 2, "JS": 8, "Img": 32, "Fetch/XHR": 0, "Font": 0,
     ↪  "Other": 2},
    "Daily Camera home page": {"Doc": 34, "CSS": 16, "JS": 127, "Img": 246, "Fetch/XHR": 60,
     ↪  "Font": 5, "Other": 9},
}
by_type = pd.DataFrame(counts)

ax = by_type.plot.bar(figsize=(9, 4), rot=0)
ax.set_title("Requests in one page load, by Network tab filter")
ax.set_xlabel("Network tab filter")
ax.set_ylabel("Number of requests")
plt.tight_layout()
plt.show()

totals = by_type.sum()
print(totals)
print((by_type.loc["JS"] / totals).round(2))
```




```
Wikipedia article          45
Daily Camera home page    497
dtype: int64
Wikipedia article         0.18
Daily Camera home page    0.26
dtype: float64
```

💬 Discuss

- Your browser made dozens of requests for the article; `requests` made one. What does a scraper never see, and what does it spare the server?
- In week 3 we read the `server` and `via` headers to learn who refused us. What does Wikimedia's `x-cache` header let you say about who answered?
- A news home page can load hundreds of requests from dozens of hosts. Who learns that you visited it?

Step 3 — Below HTTP: addresses, and one connection narrated (7 minutes)

HTTP rides on TCP/IP: every machine has an IP address, and a request travels as packets between two of them. `socket` asks your operating system for your machine's name and local address; a service like ipify reports the public address servers see. Mask its last two numbers before you print it, as the chapter does, because notebooks end up on screens and in repositories.

Then let `curl` narrate one request. A line that starts with `!` runs in your terminal, and `lines = !command` keeps its output as a list. `curl -v` narrates on its error stream, so add `2>&1` to keep it, and print only the lines for the connection (`* Connected`), the TLS handshake's result (`* SSL connection`), the request (`>`), and the response's status line (`< HTTP`). For more, see the chapter's TCP/IP: The Transport Layer and ”Packets”.

If there's time, the optional cell after it pings Wikipedia: four small packets out, four replies back. Campus Wi-Fi may drop them.

```
# Import socket. Print your hostname, and the address socket.gethostbyname() gives for it
# Get your public IP from https://api.ipify.org?format=json, and print it with its last two
 ↪  numbers replaced by xxx
# Run lines = !curl -v -I -A "<your User-Agent>" https://en.wikipedia.org/wiki/Web_scraping 2>&1
 ↪  and print the lines
#   that start with "* Connected", "* SSL connection", ">", or "< HTTP"
import socket
```

```
hostname = socket.gethostname()
print(f"Hostname: {hostname}")
print(f"Local IP: {socket.gethostbyname(hostname)}")

public_ip = requests.get("https://api.ipify.org?format=json", headers=HEADERS,
 ↪  timeout=10).json()["ip"]
print("Public IP:", ".".join(public_ip.split(".")[:2] + ["xxx", "xxx"]))

lines = !curl -v -I -A "WebDataScience/1.0 (your-email@colorado.edu)"
 ↪  https://en.wikipedia.org/wiki/Web_scraping 2>&1
for line in lines:
    if line.startswith(("* Connected", "* SSL connection", ">", "< HTTP")):
        print(line)
```

```
Hostname: vm
Local IP: 127.0.0.1
```

```
Public IP: 160.79.xxx.xxx
```

```
* Connected to 127.0.0.1 (127.0.0.1) port 35529
> CONNECT en.wikipedia.org:443 HTTP/1.1
> Host: en.wikipedia.org:443
> User-Agent: WebDataScience/1.0 (your-email@colorado.edu)
> Proxy-Connection: Keep-Alive
>
< HTTP/1.1 200 Connection Established
* SSL connection using TLSv1.3 / TLS_AES_128_GCM_SHA256 / X25519 / id-ecPublicKey
> HEAD /wiki/Web_scraping HTTP/2
> Host: en.wikipedia.org
> User-Agent: WebDataScience/1.0 (your-email@colorado.edu)
> Accept: */*
>
```

```
# Optional: ping Wikipedia four times (-n on Windows, -c on macOS and Linux)
import platform

if platform.system() == "Windows":
    !ping -n 4 en.wikipedia.org
else:
    !ping -c 4 en.wikipedia.org
```

💬 Discuss

- `socket.gethostbyname()` may give you 127.0.0.1, which means "this machine". Why is the public address the one a server's logs record?
- curl reports the connection done before it sends a byte of HTTP. Which layer did that work, and why doesn't curl show its three packets?
- Week 1 asked us to treat `requests` as a well-made black box. Which parts of what curl narrated does `requests` do for you without a word?

Step 4 — DNS: from a name to an address (8 minutes)

Before any of that, the name had to become an address. `dns.resolver.resolve()` asks a DNS server for a name's **A** records, its IPv4 addresses. The answer holds more than the address: `canonical_name` is the name the record belongs to, when the name you asked for is an alias, and `rrset.ttl` is how many seconds resolvers may cache the answer. `socket.gethostbyname()`, the lookup `requests` relies on,

should agree. If `dns.resolver` raises `NoNameservers ... answered REFUSED`, see the chapter's Common Issues to Debug. For more, see the chapter's DNS: The Address Book.

```
# Import dns.resolver. Resolve en.wikipedia.org's A record, and print each address in the answer
# Print the answer's canonical_name and rrset.ttl, and what
 ↪  socket.gethostbyname("en.wikipedia.org") says
# Keep the first address, as a string, in WIKI_IP
import dns.resolver

answers = dns.resolver.resolve("en.wikipedia.org", "A")
for answer in answers:
    print(f"IP address: {answer}")
print(f"Canonical name: {answers.canonical_name}")
print(f"TTL: {answers.rrset.ttl} seconds")
print(f"socket says: {socket.gethostbyname('en.wikipedia.org')}")
WIKI_IP = str(answers[0])
```

```
IP address: 208.80.154.224
Canonical name: dyna.wikimedia.org.
TTL: 180 seconds
socket says: 208.80.154.224
```

⏱️ Microsprint (5 minutes, alone or in pairs)

Do: find where that address is. Write the chapter's `geolocate_ip()`, from its traceroute section, which asks ip-api about an address, and look up `WIKI_IP`. Then print Step 1's `Server` header: Wikimedia names its servers after their data centers, and `eqiad` is Ashburn, Virginia; `codfw` is Carrollton, Texas. **Share:** your address, the city ip-api gives it, and the data center in your `Server` header.

```
# Microsprint: define geolocate_ip(ip), as the chapter does, but with HEADERS: request
 ↪  http://ip-api.com/json/{ip} and return the JSON
# Look up WIKI_IP and print its city, regionName, and isp. Then print response.headers["Server"]
 ↪  from Step 1
# (ip-api allows 45 requests a minute from one address, and this room shares one)
def geolocate_ip(ip):
    """Look up the geographic location of an IP address."""
    response = requests.get(f"http://ip-api.com/json/{ip}", headers=HEADERS, timeout=10)
    return response.json()

location = geolocate_ip(WIKI_IP)
print(WIKI_IP, "|", location["city"], location["regionName"], "|", location["isp"])
print(response.headers["Server"])
```

```
208.80.154.224 | San Francisco California | Wikimedia Foundation Inc.
mw-web.eqiad.main-855b9df8d-fzg98
```

💬 Discuss

- In the key, ip-api put Wikipedia's address in San Francisco, where the Wikimedia Foundation has its office, while the `Server` header named a data center in Virginia or Texas. What does an IP geolocation service look up, and which answer do you believe?
- Did everyone in the room get the same address? Why might someone on a phone hotspot get a different one?
- Resolvers may keep this answer for `ttl` seconds. If Wikipedia moved its servers tomorrow, how long could some of us still be sent to the old address?

Step 5 — Status codes, and a function that reads them (10 minutes)

HTTP is the layer you talk to. A response's status code is the server's verdict: 200 OK; 301 or 302, moved (the `Location` header says where); 403, refused; 404, not found; 429, slow down; 500, the server's own failure. Write the chapter's `diagnose_request(url, headers=None)`, which prints the status, `Content-Type`, `Server`, size, and, if `requests` followed a redirect, where from (`response.history`). Try it on the article, on a page that doesn't exist, and on the article again without `HEADERS`, to check the chapter's claim that Wikipedia answers an unnamed client with a 403. Then read the 403's body. For more, see the chapter's "Diagnosing HTTP Errors".

```
# Define diagnose_request(url, headers=None), as the chapter does: request url (timeout=10) and
 ↪  print the URL, status,
#   Content-Type, Server, size in bytes, and, if response.history isn't empty, the URL it was
 ↪  redirected from; return the response.
#   Catch requests.exceptions.ConnectionError and Timeout, and print which happened
# Run it with HEADERS on PAGE and on .../wiki/Nonexistent_Page_12345; then on PAGE with no
 ↪  headers, and print that body's first 120 characters
def diagnose_request(url, headers=None):
    """Print diagnostic information about an HTTP request."""
    try:
        response = requests.get(url, headers=headers, timeout=10)
        print(f"URL: {url}")
        print(f"Status: {response.status_code}")
        print(f"Content-Type: {response.headers.get('Content-Type', 'not specified')}")
        print(f"Server: {response.headers.get('Server', 'not specified')}")
        print(f"Response size: {len(response.content):,} bytes")
        if response.history:
            print(f"Redirected from: {response.history[0].url}")
        return response
    except requests.exceptions.ConnectionError:
        print(f"Connection failed: {url}")
    except requests.exceptions.Timeout:
        print(f"Request timed out: {url}")

diagnose_request(PAGE, headers=HEADERS)
print()
diagnose_request("https://en.wikipedia.org/wiki/Nonexistent_Page_12345", headers=HEADERS)
print()
anonymous = diagnose_request(PAGE)
print(anonymous.text[:120])
```

```
URL: https://en.wikipedia.org/wiki/University_of_Colorado_Boulder
Status: 200
Content-Type: text/html; charset=UTF-8
Server: mw-web.codfw.main-c9fb77d8f-6khgv
Response size: 768,029 bytes


URL: https://en.wikipedia.org/wiki/Nonexistent_Page_12345
Status: 404
Content-Type: text/html; charset=UTF-8
Server: mw-web.eqiad.main-855b9df8d-859p7
Response size: 50,827 bytes


URL: https://en.wikipedia.org/wiki/University_of_Colorado_Boulder
Status: 403
Content-Type: text/plain
```

```
Server: HAProxy
Response size: 126 bytes
Please set a user-agent and respect our robot policy https://w.wiki/4wJS. See also
 ↪  https://phabricator.wikimedia.org/T40
```

⏱️ Microsprint (4 minutes, alone or in pairs)

Do: follow redirects by hand. Request `https://en.wikipedia.org/` and `https://en.wikipedia.org/wiki/CU_Boulder` with `allow_redirects=False`, and print each one's status code and `Location` header. Then request `CU_Boulder` normally, and print its `response.history` and the title in its `#firstHeading`. **Share:** the two status codes, and the page each address ended up on.

```
# Microsprint: request each of the two URLs with HEADERS and allow_redirects=False; print the
 ↪  URL, status code, and Location header
# Then request the CU_Boulder URL normally: print response.url, response.history, and the text of
 ↪  its #firstHeading
for url in ["https://en.wikipedia.org/", "https://en.wikipedia.org/wiki/CU_Boulder"]:
    hop = requests.get(url, headers=HEADERS, timeout=10, allow_redirects=False)
    print(url, hop.status_code, hop.headers.get("Location"))

alias = requests.get("https://en.wikipedia.org/wiki/CU_Boulder", headers=HEADERS, timeout=10)
print(alias.url, alias.history)
print(BeautifulSoup(alias.text, "lxml").select_one("#firstHeading").text)
```

```
https://en.wikipedia.org/ 301 https://en.wikipedia.org/wiki/Main_Page

https://en.wikipedia.org/wiki/CU_Boulder 200 None

https://en.wikipedia.org/wiki/CU_Boulder []
University of Colorado Boulder
```

💬 Discuss

- The 403 came with a `Server` header and a body. In week 3 we read both before blaming a platform for a refusal. Who refused us here, and how do you know?
- `CU_Boulder` answered 200 but showed another article's title. What would a check of the status code alone have missed?
- A 404 page is still a page, with a size and a Content-Type. Why did week 2's `responsible_get()` call `raise_for_status()` instead of checking whether text came back?

Step 6 — Ask the pageviews API, politely (7 minutes)

Now an API. The Wikimedia pageviews API puts every parameter in the URL's path: project, access, agent, article, granularity, start, and end. Without a User-Agent that names you, it answers 403. When many clients share one address, as this room does, it may answer 429, with a `Retry-After` header saying how many seconds to wait. The chapter's `get_with_backoff()` waits that long, or 1, 2, then 4 seconds if the header is missing, and tries again; give its request a timeout too. Use January 2026, the chapter's window, so your numbers match the key's. For more, see the chapter's ”Making API Requests with Custom Headers”.

```
# Import time. Define get_with_backoff(url, headers=None, max_retries=3), as the chapter does
 ↪  (add timeout=30): on a 429,
#   sleep for Retry-After seconds if it's a number, otherwise 2 ** attempt, and try again; return
 ↪  the last response
# Build api_url: per-article pageviews for ARTICLE on en.wikipedia, all-access, all-agents,
 ↪  daily, 20260101 to 20260131
# Request it with no headers and print the status. Then get_with_backoff() it with HEADERS, print
 ↪  the status, raise_for_status(), keep .json() as data
import time
```

```python
def get_with_backoff(url, headers=None, max_retries=3):
    """Retry a request with exponential backoff on 429 errors."""
    for attempt in range(max_retries):
        response = requests.get(url, headers=headers, timeout=30)
        if response.status_code == 429:
            retry_after = response.headers.get("Retry-After")
            if retry_after and retry_after.isdigit():
                wait_time = int(retry_after)  # the server told us how long to wait
            else:
                wait_time = 2 ** attempt  # 1s, 2s, 4s
            print(f"Rate limited. Waiting {wait_time}s before retry...")
            time.sleep(wait_time)
        else:
            return response
    return response  # Return the last response even if still 429

api_url = (
    "https://wikimedia.org/api/rest_v1/metrics/pageviews/per-article"
    f"/en.wikipedia/all-access/all-agents/{ARTICLE}/daily/20260101/20260131"
)
print(requests.get(api_url, timeout=30).status_code)

response = get_with_backoff(api_url, headers=HEADERS)
print(response.status_code)
response.raise_for_status()
data = response.json()
print(len(data["items"]), "days; the first:", data["items"][0])
```

```
403
```

```
200
31 days; the first: {'project': 'en.wikipedia', 'article': 'University_of_Colorado_Boulder',
↪  'granularity': 'daily', 'timestamp': '2026010100', 'access': 'all-access', 'agent':
↪  'all-agents', 'views': 657}
```

💬 **Discuss**

- In week 1 we called this same API and got data back the first time. What did that request already carry that today's first one lacked?
- A 429 here counts everyone on your network, not only you. Why is waiting for `Retry-After` better than week 2's fixed `time.sleep(1)`?
- Why put a fixed date range in the URL rather than asking for ”the last 30 days”?

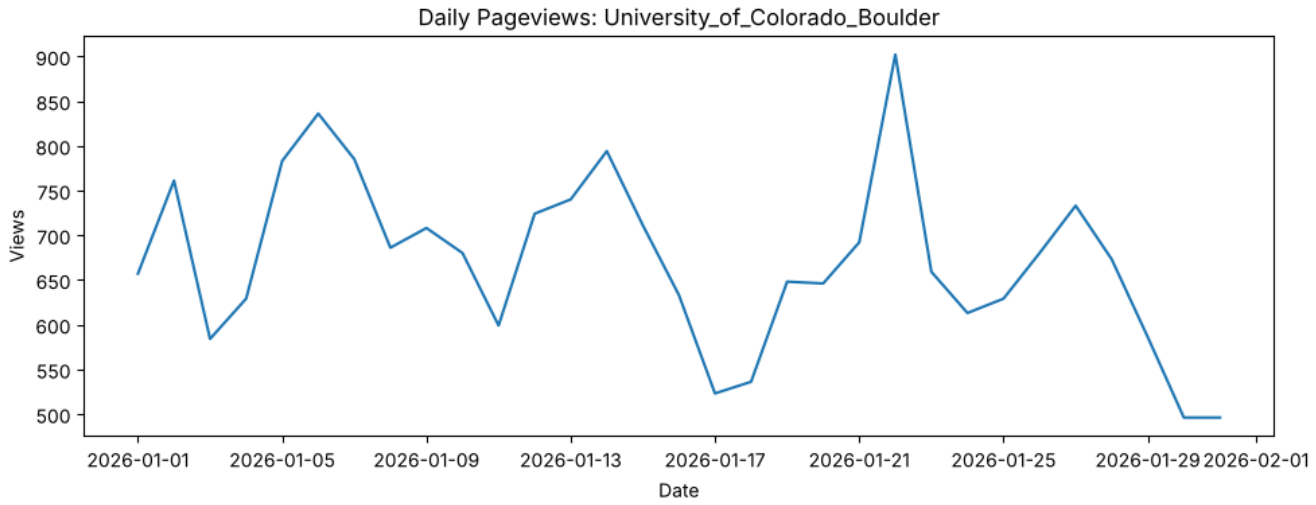
Step 7 — From JSON to a time series (9 minutes)

The views sit under `data["items"]`, one dictionary per day: week 4's list-of-dictionaries pattern, which `pd.DataFrame()` turns into a table in one call. Turn `timestamp`, written year, month, day, and hour (`2026010100`), into dates with `pd.to_datetime()` and `format="%Y%m%d00"`. Plot views by date with a title and labeled axes, and find the biggest day with `idxmax()`. For more, see the end of the chapter's ”Making API Requests with Custom Headers”.

```python
# Make a DataFrame from data["items"], and turn its timestamp into dates (format="%Y%m%d00")
# Plot views by date, with a title and labeled axes
# Print the day with the most views, and how many
df = pd.DataFrame(data["items"])
df["timestamp"] = pd.to_datetime(df["timestamp"], format="%Y%m%d00")
```

```
plt.figure(figsize=(10, 4))
plt.plot(df["timestamp"], df["views"])
plt.title(f"Daily Pageviews: {ARTICLE}")
plt.xlabel("Date")
plt.ylabel("Views")
plt.tight_layout()
plt.show()

peak = df.loc[df["views"].idxmax()]
print(peak["timestamp"].date(), f"{peak['views']:,} views")
```




```
2026-01-22 902 views
```

⏱️ Microsprint (6 minutes, alone or in pairs)

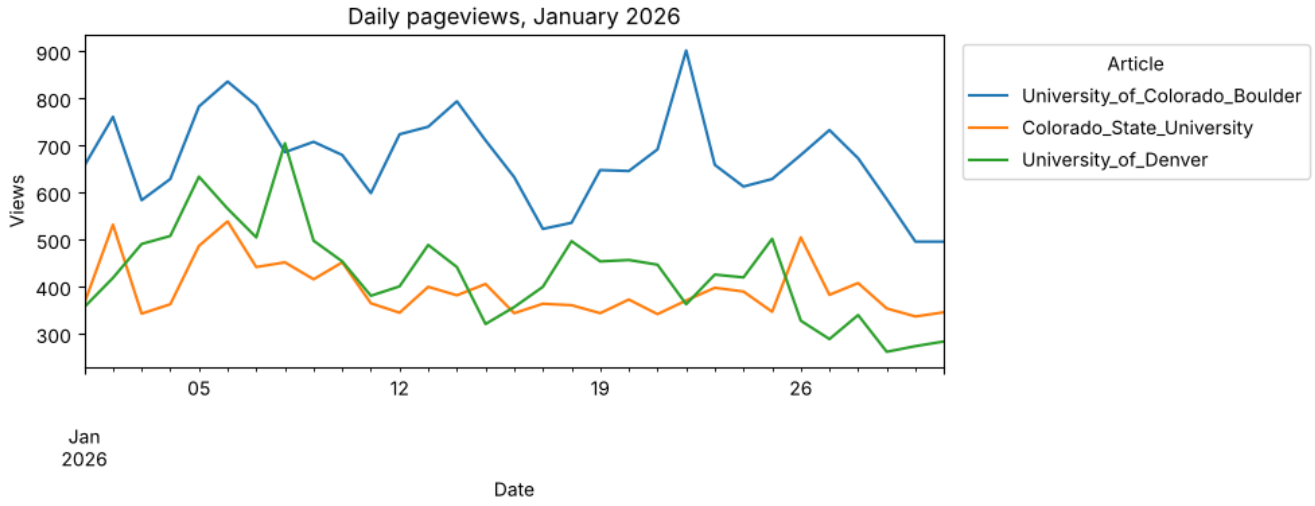
Do: a short version of Additional Exercise 3. Get the same month's daily views for two articles related to ours (another university, the city, a team), and plot all three on one set of axes. **Share:** your two articles, which of the three had the biggest spike, on what day, and a guess at why.

```
# Microsprint: for two related articles, build the same URL, get_with_backoff() it with HEADERS,
 ↪  raise_for_status(), and time.sleep(1) between
# Put all three articles' daily views in one DataFrame (one column each), and plot it with a
 ↪  title, labeled axes, and a legend
# Print each article's biggest day and its views
related = ["Colorado_State_University", "University_of_Denver"]
series = {ARTICLE: df.set_index("timestamp")["views"]}
for title in related:
    url = (
        "https://wikimedia.org/api/rest_v1/metrics/pageviews/per-article"
        f"/en.wikipedia/all-access/all-agents/{title}/daily/20260101/20260131"
    )
    reply = get_with_backoff(url, headers=HEADERS)
    reply.raise_for_status()
    views = pd.DataFrame(reply.json()["items"])
    views["timestamp"] = pd.to_datetime(views["timestamp"], format="%Y%m%d00")
    series[title] = views.set_index("timestamp")["views"]
    time.sleep(1)
```

```
compare = pd.DataFrame(series)
ax = compare.plot(figsize=(10, 4))
ax.set_title("Daily pageviews, January 2026")
ax.set_xlabel("Date")
ax.set_ylabel("Views")
ax.legend(title="Article", loc="upper left", bbox_to_anchor=(1.01, 1))
plt.tight_layout()
plt.show()

for title in compare.columns:
    print(title, compare[title].idxmax().date(), f"{compare[title].max():,} views")
```

```
Rate limited. Waiting 1s before retry...
Rate limited. Waiting 1s before retry...
Rate limited. Waiting 1s before retry...
```




```
University_of_Colorado_Boulder 2026-01-22 902 views
Colorado_State_University 2026-01-06 539 views
University_of_Denver 2026-01-08 705 views
```

💬 Discuss

- What happened on the biggest day? How would you check it, rather than guess?
- These are `all-agents` counts, crawlers included. Which claims about attention can this plot support, and which can't it?
- Week 1 plotted this article's same January. Should your numbers match the ones you got then, and why?

Step 8 — What a server sees: honest and spoofed User-Agents (4 minutes)

A `User-Agent` is whatever the client says it is. httpbin.org's `/user-agent` endpoint echoes back the header it received, so you can see exactly what a server gets: `requests`' default, your honest identification, and a **spoofed** one that claims to be desktop Chrome. Spoofing is the same dictionary with a false value. The chapter's callout explains why this course doesn't use it to get past a block. For more, see the chapter's "User-Agent Spoofing".

```
# Define spoofed_headers, whose User-Agent claims to be Chrome on Windows (copy the chapter's
 ↪  string)
# Ask https://httpbin.org/user-agent three times and print the JSON: with no headers, with
 ↪  HEADERS, and with spoofed_headers
```

```
spoofed_headers = {
    "User-Agent": (
        "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 "
        "(KHTML, like Gecko) Chrome/128.0.0.0 Safari/537.36"
    )
}
for headers in [None, HEADERS, spoofed_headers]:
    print(requests.get("https://httpbin.org/user-agent", headers=headers, timeout=10).json())
```

```
{'user-agent': 'python-requests/2.34.2'}
```

```
{'user-agent': 'WebDataScience/1.0 (your-email@colorado.edu)'}
```

```
{'user-agent': 'Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko)
 ↪  Chrome/128.0.0.0 Safari/537.36'}
```

💬 Discuss

- By this header alone, the server can't tell the spoofed request from Chrome. What else could it check? Step 2's comparison of request headers is a hint.
- Week 2 asked whether a collection is consistent with a site's terms. When is changing your User-Agent unremarkable, and when does it become getting around a block?

Step 9 — A Session that reuses its connection and retries (8 minutes)

Each bare `requests.get()` opens a connection, does the TLS handshake Step 3 narrated, and closes it. A `requests.Session` keeps the connection open for the next request to the same host, and remembers your headers. Mount an `HTTPAdapter` with a `Retry` rule, and it retries 429s and server errors for you, honoring `Retry-After`, as `get_with_backoff()` did by hand. For more, see the chapter's "Connections, Sessions, and Retries".

```
# Make a Session, put HEADERS in session.headers, and mount
 ↪  HTTPAdapter(max_retries=Retry(total=3, backoff_factor=1,
#   status_forcelist=[429, 500, 502, 503, 504])) on it for "https://"
# Loop over the chapter's three articles: get each with the session (timeout=30), and print its
 ↪  title, status, and length
from requests.adapters import HTTPAdapter
from urllib3.util.retry import Retry

session = requests.Session()
session.headers.update(HEADERS)
retries = Retry(total=3, backoff_factor=1, status_forcelist=[429, 500, 502, 503, 504])
session.mount("https://", HTTPAdapter(max_retries=retries))

articles = ["Python_(programming_language)", "Web_scraping", "Data_science"]
for article in articles:
    url = f"https://en.wikipedia.org/wiki/{article}"
    response = session.get(url, timeout=30)
    print(article, response.status_code, len(response.text))
```

```
Python_(programming_language) 200 1012849
Web_scraping 200 235982
Data_science 200 283350
```

⏱️ Microsprint (4 minutes, alone or in pairs)

Do: check the chapter's claim that a Session saves work. Time three bare `requests.get()` calls for the same three articles with `time.perf_counter()`, then three `session.get()` calls. **Share:** the two times,

and how much the Session saved.

```
# Microsprint: time three bare requests.get() calls (with HEADERS, timeout=30) for the same three
 ↪  articles, using time.perf_counter()
# Then time three session.get() calls for them, and print both times in seconds
start = time.perf_counter()
for article in articles:
    requests.get(f"https://en.wikipedia.org/wiki/{article}", headers=HEADERS, timeout=30)
bare = time.perf_counter() - start

start = time.perf_counter()
for article in articles:
    session.get(f"https://en.wikipedia.org/wiki/{article}", timeout=30)
reused = time.perf_counter() - start

print(f"bare requests.get(): {bare:.2f} s; session.get(): {reused:.2f} s")
```

```
bare requests.get(): 1.18 s; session.get(): 0.19 s
```

💬 Discuss

- Is this a fair race? The session's connection was already open from the step above. What would make the comparison fairer?
- Week 2 asked for a pause between requests. Does a Session make a loop polite on its own?
- Which costs does reusing a connection save the server, not just you?

Step 10 — Take a URL apart, and build one safely (8 minutes)

Don't parse URLs with regular expressions: `urllib.parse` knows their grammar, `scheme://host:port/path?query#fra` `urlparse()` splits a URL into its parts, and `parse_qs()` turns its query into a dictionary. Building goes the other way, in two patterns. A **path-segment** API, like pageviews, needs each value made safe for its slot with `quote(value, safe="")`. A **query-parameter** API, like the MediaWiki Action API, takes a dictionary, which `urlencode()` escapes, or which `requests` escapes for you when you pass it as `params=`. For more, see the chapter's URLs: Anatomy and Parsing.

```
# From urllib.parse import urlparse, parse_qs, quote, and urlencode. urlparse() the chapter's
 ↪  Census URL: print its scheme, host, path, and parse_qs(query)
# quote() "University of Colorado Boulder" for a path segment: spaces to underscores first, and
 ↪  safe=""
# urlencode() the chapter's Action API params (action=query, format=json, titles, prop=info);
 ↪  then session.get() them with params= and print response.url and the status
from urllib.parse import urlparse, parse_qs, quote, urlencode

census_url =
 ↪  "https://api.census.gov/data/2022/acs/acs5?get=NAME,B01001_001E&for=place:07850&in=state:08"
parsed = urlparse(census_url)
print(f"Scheme: {parsed.scheme}")
print(f"Host: {parsed.netloc}")
print(f"Path: {parsed.path}")
print(parse_qs(parsed.query))

print(quote("University of Colorado Boulder".replace(" ", "_"), safe=""))

base_url = "https://en.wikipedia.org/w/api.php"
params = {"action": "query", "format": "json", "titles": "University of Colorado Boulder",
 ↪  "prop": "info"}
print(urlencode(params))
response = session.get(base_url, params=params, timeout=30)
```

```
print(response.url)
print(response.status_code)
```

```
Scheme: https
Host: api.census.gov
Path: /data/2022/acs/acs5
{'get': ['NAME,B01001_001E'], 'for': ['place:07850'], 'in': ['state:08']}
University_of_Colorado_Boulder
action=query&format=json&titles=University+of+Colorado+Boulder&prop=info
https://en.wikipedia.org/w/api.php?action=query&format=json&titles=University+of+Colorado+Boulde
↪  r&prop=info
200
```

⏱️ Microsprint (4 minutes, alone or in pairs)

Do: check the chapter's warning about slashes. Build January 2026's pageviews URL for the article `AC/DC` twice: once with the raw title in the f-string, once with `quote(title, safe="")`. Request both with your session. **Share:** the two status codes, and what the server took the raw slash to mean.

```
# Microsprint: for the title "AC/DC", build January 2026's daily pageviews URL twice: with the
↪  raw title, and with quote(title, safe="")
# Request both with session.get(), and print the part of each URL's path after "all-agents/", and
↪  the status code
title = "AC/DC"
for segment in [title, quote(title, safe="")]:
    url = (
        "https://wikimedia.org/api/rest_v1/metrics/pageviews/per-article"
        f"/en.wikipedia/all-access/all-agents/{segment}/daily/20260101/20260131"
    )
    reply = session.get(url, timeout=30)
    print(urlparse(url).path.split("all-agents/")[1], reply.status_code)
```

```
AC/DC/daily/20260101/20260131 404
```

```
AC%2FDC/daily/20260101/20260131 200
```

💬 Discuss

- `quote("AC/DC")` without `safe=""` leaves the slash alone. Why is that the default, and why is it wrong for a path segment?
- Look at the Action API's `response.url`. Which characters did `requests` escape, and how?
- Week 3's access matrix probed APIs of both kinds. How can you tell from a URL alone which kind you have? Name one of each from that list.

Step 11 — What did we learn? (5 minutes)

In a few sentences, as comments or in a Markdown cell:

- In Step 2 and its microsprint, how many requests did your browser make for the article, and for the news site? What does a scraper that sends one request never see?
- Which of today's outputs (Steps 3 and 4 above all) describe your own network or machine rather than Wikipedia?
- Which status codes did we meet in Steps 5 and 6, and which response would a check of the status code alone have read wrongly?
- Why did the raw `AC/DC` URL in Step 10's microsprint fail, and what told you?

Recommended Exercises ahead (in the chapter's companion notebook, due Sunday):

- Step 1, take a URL apart: Step 10's `urlparse()` of the Census URL.

- Step 2, resolve a name: Step 4's `dns.resolver.resolve()`.
- Step 3, request with named headers: Step 0's `HEADERS`, and Step 2's reading of response headers (and why `Content-Length` and `len(response.content)` differ).
- Step 4, write `diagnose_url()`: Step 5's `diagnose_request()`, and its microsprint's redirects (`Location`, `response.history`).
- Step 5, profile five URLs: Steps 5 and 6 (a page, a 404, a refusal, an API), and week 3's habit of reading a refusal's `Server` header and body.
- Step 6, interpret: Step 2's compressed and decompressed sizes, Step 5's `Content-Type` and `Server` headers, and week 3's question of who answered.

```
# Your answers here, as comments
# Step 2: the key's page loads made 45 requests for the article and 497 for the Daily Camera's
 ↪  home page
#   (26% of them scripts); requests made 1, and got 768,062 bytes of HTML, sent as 121,506
 ↪  gzipped bytes.
#   A scraper never sees the scripts, images, and Fetch/XHR data the browser loads after the
 ↪  HTML.
# Our network, not Wikipedia: the local IP 127.0.0.1, the public IP (the proxy's, in the key),
 ↪  curl's
#   CONNECT to the proxy, and the DNS answer (208.80.154.224, cacheable for 180 s), which depends
 ↪  on where you
#   ask from. ip-api's "San Francisco" describes who registered the address, not where the server
 ↪  is.
# We met 200, 301 (en.wikipedia.org/ to Main_Page), 403 (HAProxy, text/plain, 126 bytes, about
 ↪  our missing
#   User-Agent), 404 (a 50,827-byte page), and 429s that get_with_backoff() waited out.
 ↪  CU_Boulder answered
#   200 with another article's title: a status-only check would read it as the page we asked for.
# The raw "AC/DC" put a slash in the path, so the API read two segments and answered 404;
#   quote(title, safe="") sent AC%2FDC and got 200. The status code told us; nothing else warned
 ↪  us.
```

Friday: from the lab to the textbook (15 minutes)

Did anything in this lab disagree with chapter 5? Places to look:

- **Step 4's microsprint.** The chapter maps traceroute hops with `geolocate_ip()`, but ip-api placed Wikipedia's address in San Francisco, at the Foundation's office, while the `Server` header named a data center in Virginia or Texas. Should the chapter say what an IP geolocation service looks up?
- **The helpers.** The chapter's `geolocate_ip()` sends no User-Agent, and `get_with_backoff()` sets no timeout, though chapter 2 asks for both. We added them here.
- **Outputs that have moved.** The chapter's Network tab figure counts 37 requests for the article, and its `Session` example prints page lengths from when it was written. What did you get?
- **A cell that broke for you,** such as `ping` on campus Wi-Fi, or `dns.resolver` off campus. Is the chapter's Common Issues to Debug enough?
- **A microsprint that would make a good exercise,** such as Step 10's `AC/DC` check.

`handouts/common/revision-framework.md` says how to propose a revision.